Fiscal Year:	FY 2004	**Task Last Updated:**	FY 09/01/2004
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Project Title:	Full-field Interferometric Measurements of Thickness of Freely-suspended Liquid Films		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid transport phenomena		
Joint Agency Name:		**TechPort:**	No
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
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Flight Program:			
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Key Personnel Changes/Previous PI:	Wu, X.L.		
COI Name (Institution):			
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Task Description:	We propose to develop interferometric methods to visualize and measure the film thickness in a two-dimensional flowing soap film. Since the film used can be sometimes several optical wavelengths in thickness, enumeration of the interference fringes resulting from a single wavelength in an interferogram is ambiguous and cannot be used to determine the precise value of the thickness. This difficulty can be overcome if interferograms are acquired simultaneously using several different wavelengths or in combination with infrared absorption measurements. The full-field thickness information on flowing liquid films is useful for understanding stress distribution in the film and how this stress relaxes over time and distance. The technique is also useful for understanding the interaction between turbulence and thickness fluctuations, which under certain conditions can be treated as a passive scalar similar to a dye or a temperature distribution. Build a flowing soap film channel that allows independent control of the film thickness and flow speed. Simple fluid structures, such as a single vortex or a few vortices created by a bluff body will be examined first. More complicated structures, such as a large array of vortices will be studied afterwards. Until recently		

	most of studies of two-dimensional turbulence in flowing soap films have been concentrated on velocity fluctuations. However, our experience shows that accompanying the velocity fluctuations, the thickness of the film also fluctuates. The dynamic role of the thickness variation is not clear at this point. Theoretically, it is predicted that for the typical velocity of the flow to be least than the sound velocity in the film, the thickness variation is passively convected. On the other hand, when the typical velocity of the flow is greater than the sound velocity, the shock waves are created and the thickness variation is no longer passive. The two-dimensional experiment proposed here allows us to investigate the intricate coupling between the velocity field and the flow medium. Similar problems occur naturally in oceans and in atmosphere.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Progresses are made over the past year to develop a radiation thermometer to detect thermal radiations and thickness fluctuations in a flowing soap film. The technique relies on thermal imaging, a well-developed analytical tool. However, measuring thermal radiation from a thin soap film a few microns thick is a different matter because the thermal mass is so small such that unless a very sensitive detection method is implemented, the measurements would be contaminated by noises. To measure tiny radiation signal from the film we used a mercury-cadmium-tellurium (HgCdTe) detector manufactured by EG&G. The detector was protected by a double-window Dewar and cooled down to the liquid nitrogen temperature of 77K. To test the system we employed a freely suspended soap film that was driven by a vertical temperature gradient similar to the classical Rayleigh-Benard convection. The novel feature of the experiment is that the film thickness is very thin and flow is confined to the plane of the film, behaving two dimensionally. We found that for low temperature gradients, turbulence is weak and is confined to the lower part of the convection cell. As the temperature gradient increases, turbulent intensity increases and eventually the entire cell is well mixed by turbulent fluid flows. This partial mixing of the fluid is new in this experiment and indicates that density stratification plays the crucial role in turbulent mixing, which is very different from Rayleigh-Benard convection. A back-of-the-envelop calculation shows that a film of 2 cm in height is equivalent to 7 km in the earth's atmosphere measured from the sea level, making our system a unique tabletop experiment to mimic thermal convection in lower atmosphere. We have developed a systematic method to calibrate the emitted thermal power and its temporal fluctuations at a local point in the film. Our measurements yield useful information about local temperature and thickness variations as a function of time. We showed that for an IR detector with a spectral range of 8-14 mm, the dominant temporal contribution to the radiation signal is due to fluctuations of the thermal emissivity of the film, which is a function of the film thickness h. The technique thus provides a rapid and convenient means to measure mass density fluctuations in the film. A variance of this technique allows direct imaging of the film thickness and reveals for the first time the correlation between the formation of a large-scale circulation and the asymmetric mass distribution in the convection cell. Our experiment also indicates that it is feasible to measure temperature fluctuations in the film if the detector is operated at near 3-mm wavelength, where water absorbs the electromagnetic waves most strongly with an absorption length of less than one micrometer. In this strong absorption regime, a film of a few microns in thickness is essentially opaque, behaving like an ideal blackbody with emissivity close to unity. The radiated thermal power should in principle allowing direct measurements of local temperature in the film. The above work was presented in a recent American Physical Society Meeting in Montreal, Canada, and is under review by Physical Review E. Using the same technique, we also studied physics in strongly stratified liquid media, a subject that is of interest to NASA. In a closed system such as our soap films, when the kinetic energy (KE) of thermal plumes becomes greater than gravitational potential energy (PE) of stratification, a large-scale circulation (LSC) takes place and occupies the entire cell. This new state of motion is very robust in that it has a long persistent time and is unsusceptible to external disturbances. The presence of the LSC significantly alters the mass transport in the convection cell and can be quantitatively measured by the single-point IR detector we developed. This work has just been completed and is also under review by Physical Letters.
Bibliography Type:	Description: (Last Updated: 05/08/2019)
Articles in Peer-reviewed Journals	Jie Zhang, X.L. Wu, N. Rashidnia "Thermal Radiation and Thickness Fluctuations in Freely Suspended Liquid Films" Phys. Rev. E, under review , Mar-2004
Articles in Peer-reviewed Journals	Zhang, Jie; Wu, X.L.; and Xia, K.Q. "Density Fluctuations in Strongly Stratified Two-Dimensional Fluids" under review , Aug-2004